MILITARY SPECIFICATION

RADIO SET AN/GRC-109( )(UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of radio set, designated Radio Set AN/GRC-109( ):

Radio Receiver R-1004( )/GRC-109
Radio Transmitter T-784( )/GRC-109
Power Supply PP-2684( )/GRC-109
Power Supply PP-2685( )/GRC-109

The set is a compact, portable radio station used for general purpose continuous wave communications over the frequency range of 3 to 22 megacycles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 —Boxes, Wood, Wirebound.
PPP-B-591 —Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 —Boxes, Wood, Cleated Plywood.
PPP-B-621 —Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 —Box, Fiberboard.
PPP-T-97 —Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-E-1 —Electron Tubes and Crystal Rectifiers
MIL-C-104 —Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 —Preservation, Methods of.
MIL-C-132 —Crate, Wood, Open, Maximum Capacity 2,500 pounds.
MIL-V-173 —Varnish, Moisture and Fungus-Resistant.
MIL-T-713 —Twine and Tape, Lacing and Tying (For Use in Electrical and Electronic Equipment).
MIL-C-3774 —Crates, Open, Wood (2,500 to 10,000 lb.).
MIL-B-10377 —Box, Wood, Cleated, Veneer, Paper-Overlaid.
MIL-I-11748 —Interference Reduction for Electrical Equipment.
MIL-M-13231 —Marking of Electronic Items.
MIL-F-14072 —Finishes for Ground Signal Equipment.
*MIL-G-14106 —Generator, Direct Current, G-43( )/G (Hand-Cranked).
MIL-C-45662 —Calibration of Standards.

STANDARDS

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

* For informational purposes only (see 3.17.1).

FSC-5820

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MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-169 —Extreme - Temperature Cycle.
MIL-STD-170 —Moisture Resistance Test Cycle, for Ground Signal Equipment.
MIL-STD-252 —Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-A-362100 —Preparation of Preproduction Packs.
SC-DL-433000 —Power Supply PP-2685( )/GRC-109.
SC-DL-433149 —Radio Transmitter T-784( )/GRC-109.
SC-DL-433299 —Radio Receiver R-1004( )/GRC-109.
SC-DL-433623 —Power Supply PP-2684( )/GRC-109.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. The principal units of the radio set, which provide communication under a wide range of climatic conditions, are a low power transmitter, a superheterdyne receiver, and two power supplies. The two power supplies permit operation from a variety of primary power sources and a wide range of input voltages. In addition to the units covered by this specification, Voltage Regulator CN-690( )/GRC-109, Headset H-65( )/U, cable assembly, and a single wire antenna are part of the radio set.

3.2 Preproduction samples. The contractor shall furnish two preproduction samples of each unit on order for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2(e)).

3.3 Construction. The units of Radio Set AN/GRC-109( ) covered by this specification shall be constructed in accordance with the applicable drawing and data list indicated below:

Unit	*Drawing and data list*
Radio Receiver R-1004( )/GRC-109	SC-DL-433299
Radio Transmitter T-184( )/GRC-109	SC-DL-433149
Power Supply PP-2684( )/GRC-109	SC-DL-433623
Power Supply PP-2685( )/GRC-109	SC-DL-433000

3.4 Welding. Welding shall conform to the equipment drawings, and shall be such that grinding on the finished weld will be unnecessary.

3.4.1 *Cleaning prior to welding.* Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.4.2 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.4.3 *Cleaning after welding.* Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means.

3.5 Tube, electron. Tubes shall conform to Specification MIL-E-1 (see 4.4).

3.6 Cleaning.

3.6.1 *Parts.* After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Crossive material shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.6.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.7 Controls. All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misalined condition.

3.8 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (see 4.4). The final paint film on Type I surfaces shall be green color (olive-drab) enamel matching a color chip provided by the procuring agency (see 6.3).

3.9 Marking.

3.9.1 *Conformance.* Marking shall conform to Specification MIL-M-13231 (see 4.4.). Front panel markings shall be group I as described in that specification and the equipment drawings.

3.9.2 *Visibility.* Wherever practicable, parts shall be mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9.3 *Serial numbers.* Radio Receiver R-1004( )/GRC-109, Radio Transmitter, T-784( )/GRC-109, Power Supply PP-2684( )/GRC-109, and Power Supply PP-2685( )/GRC-109 shall have serial numbers.

3.10 Riveting. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.11 Securing of parts.

3.11.1 *Mounting of items.* Brakets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.11.2 *Securing of parts by threaded fasteners.* Nylok or other screws with plastic devices, and Loctite or similar sealants shall not be subjected to temperatures in excess of 250° F. (as during baking of paint). Loctite or similar sealants shall not be used where the connection must be electrically conductive; where used, such materials shall be applied in accordance with the manufacturer's instructions.

3.11.3 *Brittle material.* Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Not more than three points of contact shall be used between the brittle part and the mounting surface, unless the mating faces are accurately machined. Mounting washers of suitable plastic, rubber, or soft copper having slight compressibility shall be provided to prevent breakage or cracking of the parts. Lead washers shall not be used. Threaded holes in ceramic material shall not be employed for assembly or mounting of parts.

3.12 Soldering.

3.12.1 *Solder.* Solder shall conform to the equipment drawings.

3.12.2 *Acid or acid salts.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.12.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment: flux being spread over a larger area; or damage to the equipment. Insulation material that

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has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.13 **Wiring and cabling.** Wiring and cabling shall be neat and sturdy.

3.13.1 *Cabling.* Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.13.2 *Slack.* Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

(d) To insure freedom of motion of lugs or terminals normally intended to have some degree of movement.

(e) To facilitate field repair of broken or cut wires.

(f) To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.13.3 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.13.4 *Support.* Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire and cable.

3.13.5 *Clearance.* Clearance between wires or cables, and parts shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.13.6 *Splicing.* Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.13.7 *Connections.* Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.14 **Service conditions.** The equipment shall meet the following service conditions—where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.14.1 *Operation.* All units shall be capable of continuous operation for a period of one year under the worldwide environmental con-

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ditions specified in 3.14.2 through 3.14.8, with no more than normal maintenance and replacement of parts. The duty cycle for Radio Receiver R-1004( )/GRC-109, Power Supply PP-2684( )/GRC-109, and Power Supply PP-2685( )/GRC-109 shall be 23 hours on and one hour off. The duty cycle for Radio Transmitter T-784( )/GRC-109 shall be two hours on and two hours off. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.14.2 *Temperature (see 4.12.1).*

(a) *Operating.* Ambient temperature in range of +150° F. to −65° F. (The 150° F. temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) *Nonoperating.* Exposure in the range of +160° F. to −80° F.; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.14.3 *Relative humidity (see 4.12.2).* Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.14.4 *Elevation (see 4.12.3).*

(a) *Operating.* Up to 10,000 feet above sea level.

(b) *Nonoperating.* Up to 50,000 feet above sea level.

3.14.5 *Immersion (see 4.12.4).* Three feet of water for 2 hours.

3.14.6 *Orientation (see 4.12.5).*

(a) *Operating.* Any orientation up to 20 degrees from normal operating position (forward, backward, left, or right).

(b) *Nonoperating.* Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample shall be considered as compliance with this requirement.

3.14.7 *Rain (see 4.12.6).* Two inches of rainfall per hour, with wind at 20 miles per hour.

3.14.8 *Sand and dust (see 4.12.7).* As encountered during heavy traffic over dry clay roads or in desert areas.

3.15 Electrical requirements, Radio Receiver R-1004( )/GRC-109.

3.15.1 *Frequency range (see 4.8.1).* The receiver shall provide continuous frequency coverage from 3,000 to 24,000 kilocycles. The frequencies are covered in three bands, as follows:

Band	*Frequency range*
1	3,000 to 6,000 kc
2	6,000 to 12,000 kc
3	12,000 to 24,000 kc

3.15.2 *Sensitivity (see 4.8.2).* Radio frequency signal inputs of 1.5 microvolts shall produce a signal plus noise to noise ratio of at least 10 decibels (db) with the receiver adjusted for AM reception.

3.15.3 *IF selectivity.* The intermediate frequency selectivity of the receiver shall be within the following limits when measured in accordance with 4.8.3:

Reference signal—db down	*Maximum bandwidth—kc.*
6	9
20	13
40	20
60	30

3.15.4 *Image rejection (see 4.8.4).* The image rejection ratio shall be not less than 34 db down.

3.15.5 *Spurious responses (see 4.8.5).* The rejection ratio of all spurious responses except the image rejection ratio shall exceed 60 db down.

3.15.6 *IF rejection (see 4.8.6).* The IF rejection ratio shall be not less than 100 db down.

3.15.7 *Cross signal distortion (see 4.8.7).* The amplitude of cross signal distortion

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products shall be more than 50 db down from the desired signal.

3.15.8 *Dial calibration (see 4.8.8).* The dial calibration error shall be less than 0.275 percent.

3.15.9 *Resettability (see 4.8.9).* There shall be a difference of not more than ½-vernier division in the setting of the dial when the test frequency is approached from the clockwise and counter-clockwise direction.

3.15.10 *B.F.O. tuning range (see 4.8.10).* The beat frequency oscillator shall have a minimum tuning range of ±4 kc.

3.15.11 *Frequency stability with warm-up (see 4.8.11).* The final frequency, after three hours of operation, shall not deviate by more than 0.02 percent from the frequency obtained after a 15 minute warm-up period.

3.15.12 *Plate voltage range (see 4.8.12).* The receiver output shall not vary more than 3 db from that obtained with plate voltage of 100 volts when the plate voltage is varied from 95 to 110 volts.

3.15.13 *Overall audio frequency responses (see 4.8.13).* The overall audio frequency response shall be within ±4.5 db of the 1000 cps reference from 300 to 1600 cps. The response above 1600 cps shall fall so as to be down a minimum of 10 db at 3000 cps.

3.15.14 *Audio power output (see 4.8.14).* The audio power output shall be a minimum of 30 milliwatts into a 4000-ohm load across the headphone terminals.

3.15.15 *Crystal operation (see 4.8.15).* Insertion of the proper crystal in the front panel shall permit operation as a fixed frequency receiver at any frequency in the tuning range. The sensitivity shall be 1.5 microvolts or better on fundamental and second harmonic operation, and 3.0 microvolts on third harmonic operation.

3.16 Electrical requirements, Radio Transmitter T–784( )/GRC–109.

3.16.1 *Frequency range (see 4.9.1).* The transmitter shall provide continuous frequency coverage from 3,000 to 22,000 kc. The frequencies are covered in four bands, as follows:

Band	*Frequency range*
1	3,000 to 6,000 kc.
2	6,000 to 10,000 kc.
3	10,000 to 17,000 kc.
4	17,000 to 22,000 kc.

3.16.2 *Power output (see 4.9.2).* The radio frequency power output shall be a minimum of 10 watts for all frequencies except 5.5 megacycles in band 1 where the minimum shall be 5 watts.

3.16.3 *Keying characteristics (see 4.9.3).* Keying characteristics shall be clean and free from transients. There shall be no frequency shift with keying.

3.16.4 *Keying speed (see 4.9.4).* The transmitter shall be capable of keying speeds up to 300 wpm using the AN/GRA–7( ) Coder-Burst Transmission Group.

3.17 Electrical requirements, Power Supply PP–2684( )/GRC–109.

3.17.1 *Power sources (see 4.10.1).* The unit shall be capable of operation from the following power sources:

75 to 260 volts a.c., 40 to 400 cycles single phase.
6 volt storage battery
Generator, Direct Current, G–43( )/G (Hand-Cranked) MIL–G–14106

3.17.2 *Power outputs (see 4.10.2).* The unit shall provide the following simultaneous outputs using any of the specified input voltages:

450 ±25 volts at 100 ma.
100 ±10 volts at 20 ma.
1.35 ±0.1 volts at 300 ma.
6.3 ±0.4 volts at 1.5 amps.

3.17.3 *Ripple voltage (see 4.10.3).* The ripple voltage shall be less than 5 volts for the 450 volt output, less than 0.04 volt for the 100 volt output, and less than 0.04 volt for the 1.35 volt output.

3.17.4 *Noise output (see 4.10.4).* No vibrator hash noise shall be detected in Radio

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Receiver R-1004( )/GRC-109 when the power supply is operated from a 6 volt battery.

3.17.5 *Battery charging output (see 4.10.5)*. The unit, operating on 110 volts a.c. input, shall provide a direct current output of 2 amperes or more at 7 volts, suitable for charging a 6 volt storage battery.

3.18. Electrical requirements, Power Supply PP-2685( )/GRC-109.

3.18.1 *Power outputs (see 4.11.1)*. The unit shall provide the following simultaneous outputs with inputs of 75 to 260 volts a.c., single phase, 40 to 400 cycles:

450 ±25 volts at 100 ma.
100 ±10 volts at 20 ma.
1.35 ±0.1 volts at 300 ma.
6.3 ±0.4 volts at 1.5 amps.

3.18.2 *Ripple voltage (see 4.11.2)*. The ripple voltage shall be less than 10 volts for the 450 volt output, less than 0.04 volt for the 100 volt output, and less than 0.04 volt for the 1.35 volt output.

3.19 Preconditioning.

3.19.1 *Bounce preconditioning*. The units shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1 (also see 4.5).

3.19.2 *Electrical preconditioning*. All units on order shall be electrically preconditioned for one half hour without degradation of sensitivity or power output (see 4.5 and 4.6.2).

3.20 Air-seal test. The decrease in vacuum shall not exceed 0.01 pounds per square inch when the units of this equipment are tested in accordance with 4.13.

3.21 *Operational inspection*. The operational test specified in 4.14 shall be performed to determine that the equipment is operable prior to packaging and at such other times as required.

3.22 500 hour operational requirement. All units shall be capable of continuous operation for a period of 500 hours with no replacement of parts or realignment other than adjustment of front panel controls. The duty cycle for each unit shall be as follows:

Radio Receiver R-1004( )/GRC-109—continuous
Radio Transmitter T-784( )/GRC-109—two hours on and two hours off using on-off keying
Power Supply PP-2684( )/GRC-109—continuous
Power Supply PP-2685( )/GRC-109—continuous

At the conclusion of operation, the radio set shall meet the tests specified in 4.16.

3.23 Interchangeability. Like units, assemblies, subassemblies, modules, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.15). Individual items shall not be hand-picked, for fit or performance; however, matched pairs or sets when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 Suppression radio interference (see 4.4). The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class I, with the exceptions thereto specified in the bid request and contract.

3.25 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.26 Workmanship. The equipment shall be manufactured and assembled in accordance with the equipment drawings and the applicable portions of the following paragraphs:

3.4 Welding
3.6 Cleaning
3.7 Controls
3.10 Riveting

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3.11 Securing of parts
3.12 Soldering
3.13 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilty for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Test and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.18).

4.3 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables II, III, and IV respectively).

TABLE I. *Preproduction inspection*

Inspection	Requirements paragraph	Inspection paragraph
Temperature	3.14.2	4.12.1
Moisture resistance	3.14.3	4.12.2
Altitude	3.14.4	4.12.3
Orientation	3.14.6	4.12.5
Sand and dust	3.14.8	4.12.7
Rain	3.14.7	4.12.6

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	*Where required*
Tube, electron	3.5
Finish	3.8
Marking	3.9
Suppression	3.24

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.10). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection, except preproduction samples, shall be preconditioned after final assembly (see 3.19).

4.5.1 *Group A inspection.* This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government except the

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TABLE II. *Group A inspection*

Inspection	Requirements paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical:	3.26	4.17		
Radio Receiver R-1004 ( )/GRC-109			1.5 dphu	6.5 dphu
Radio Transmitter T-784( )/GRC-109			1.5 dphu	6.5 dphu
Power Supply PP-2684( )/GRC-109			1.5 dphu	6.5 dphu
Power Supply PP-2685( )/GRC-109			1.5 dphu	6.5 dphu
Electrical:				
Radio Receiver R-1004( )/GRC-109:				
Frequency range	3.15.1	4.8.1	4.0%	(*)
Sensitivity	3.15.2	4.8.2		
IF selectivity	3.15.3	4.8.3		
Dial calibration	3.15.8	4.8.8		
Plate voltage range	3.15.12	4.8.12		
Overall audio frequency response	3.15.13	4.8.13		
Audio power output	3.15.14	4.8.14		
Crystal operation	3.15.15	4.8.15		
Radio Transmitter T-784( )/GRC-109:				
Frequency range	3.16.1	4.9.1	4.0%	(*)
Power output	3.16.2	4.9.2		
Keying characteristics	3.16.3	4.9.3		
Keying speed	3.16.4	4.9.4		
Power Supply PP-2684( )/GRC-109:				
Power outputs	3.17.2	4.10.2	4.0%	(*)
Ripple voltage	3.17.3	4.10.3		
Noise output	3.17.4	4.10.4		
Battery charging rate	3.17.5	4.10.5		
Power Supply PP-2685( )/GRC-109:				
Power outputs	3.18.1	4.11.1	4.0%	(*)
Ripple voltage	3.18.2	4.11.2		
Air seal	3.20	4.13	1.0%	(*)
Operational	3.21	4.14	1.0%	(*)

* Defects in these categories are considered major.

air-seal test (4.13) shall be next to last and the operational inspection (4.14) shall be last (see 6.7).

4.5.2 *Group B inspection.* This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5% defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 *Group C inspection.* This inspection shall be as listed in Table IV and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

4.5.3.1 *Sampling for inspection of equipment.* Two units of each type on order shall be selected from the first quantities produced,

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TABLE III. *Group B inspection*

Inspection	Requirements paragraph	Inspection paragraph	AQL (See 4.5.2)
Radio Receiver R-1004 ( )/GRC-109:			
Image rejection	3.15.4	4.8.4	
Spurious response	3.15.5	4.8.5	
IF rejection	3.15.6	4.8.6	
Cross signal distortion	3.15.7	4.8.7	
Resettability	3.15.9	4.8.9	6.5%
B.F.O. tuning range	3.15.10	4.8.10	
Frequency stability	3.15.11	4.8.11	
Interchangeability	3.23	4.15	6.5%
Immersion (all units)	3.14.5	4.12.4	6.5%

GROUP IV. *Group C inspection*

Inspection	Requirements paragraph	Inspection paragraph
500 hour operation	3.22	4.16

thereafter two units of each type from each 200, or fraction thereof produced, or two units from each month's production, whichever occurs first.

4.5.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.6).

4.6 Preconditioning.

4.6.1 *Bounce preconditioning.* The units of Radio Set AN/GRC-109( ) shall be placed in their normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The units shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the units shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection (see 3.19.1).

4.6.2 *Electrical preconditioning.* All units shall operate continuously for a period of ½ hour, except the transmitter, which shall operate with on-off keying for the same period (see 3.19.2).

4.7 Standard test conditions. Standard test conditions are those which are to be maintained for the particular performance requirements described herein except where otherwise specified for performing a particular test:

Temperature	Room ambient
Humidity	Room ambient
Warm-up time	15 minutes
Input voltage	110 V AC 60 cycles
Receiver load impedance	4000 ohms
Receiver dummy antenna	270 ohms
Receiver gain control	⅓ maximum
Beat frequency oscillator	off
Signal generator modulation	30 percent at 400 cycles

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4.8 Receiver electrical tests.

4.8.1 *Frequency range.* The signal generator without modulation shall be connected to the receiver antenna post. The receiver shall be tuned to the lowest frequency in band 1 with the BFO turned on and set to its center position. The signal generator shall be tuned to produce zero beat and the frequency measured with a frequency counter. The procedure shall be repeated with the receiver tuned to the highest frequency in band 1. With the band switch set for band 2 and band 3 the procedure shall be repeated (see 3.15.1).

4.8.2 *Sensitivity.* The receiver shall be adjusted for standard test conditions except the gain control shall be set at two thirds of maximum. The signal generator output shall be 1.5 microvolts modulated. The signal plus noise to noise ratio shall be measured at 3.0 and 6.0 megacycles in band 1, at 6.0 and 12.0 megacycles in band 2, and at 12.0 and 24 megacycles in band 3 to determine compliance with 3.15.2.

4.8.3 *IF selectivity.* The signal generator with modulation shall be connected to pin 6 of the receiver converter tube through a 0.1 microfarad capacitor and shall be tuned to the IF center frequency. The input signal from the generator shall be adjusted to produce a reference output of 2.5 milliwatts. The signal input shall be successively increased to two, ten, one hundred, and one thousand times its initial value. At each value the signal generator shall be detuned to each side of the center frequency to the points where the reference output is obtained (see 3.15.3).

4.8.4 *Image rejection.* The signal generator output with modulation shall be adjusted to 10 microvolts at each test frequency to establish a reference output. The image rejection ratio shall be measured at 6 mc in band 1, 12 mc in band 2, and 24 mc in band 3 (see 3.15.4).

4.8.5 *Spurious response.* The signal generator output with modulation shall be adjusted to 10 microvolts at each test frequency to establish a reference output. The signal generator output shall then be increased to 10,000 microvolts and slowly tuned through the frequency range of 0.5 to 30 megacycles. The test shall be performed at 3, 4.5, and 6 megacycles in band 1; at 6, 9, and 12 megacycles in band 2; and at 12, 18, and 24 megacycles in band 3 (see 3.15.5).

4.8.6 *IF rejection.* The signal generator output with modulation shall be adjusted to 10 microvolts at each test frequency to establish a reference output. The signal generator shall then be tuned to the IF frequency and the output increased to produce the reference output. The test shall be performed at 6 megacycles in band 1, at 12 megacycles in band 2, and at 24 megacycles in band 3 (see 3.15.6).

4.8.7 *Cross signal distortion.* The receiver shall be adjusted for standard test conditions except the gain control shall be at minimum. A signal generator with modulation shall be adjusted to 10 microvolts output at each test frequency to produce a reference output. A second generator with modulation shall be connected to the receiver and tuned to 0.25 megacycles. The receiver shall be successively tuned to the sum and difference frequencies. The output of both generators shall be kept equal and simultaneously increased until the reference output is produced. This test shall be performed at 5.7, 11.7, and 23.7 megacycles (see 3.15.7).

4.8.8 *Dial calibration.* The receiver shall be adjusted for standard test conditions except the beat frequency oscillator shall be on. The tuning dial shall be successively set to 3.0, 3.6, 4.8, and 6.0 megacycles in band 1; 6.0, 7.8, 9.8, and 12 megacycles in band 2; and 12.0, 15.6, 20.4, and 24.0 megacycles in band 3. At each test point a frequency counter shall be used to measure the frequency of a signal generator tuned to the receiver frequency (see 3.15.8).

4.8.9 *Resettability.* The receiver shall be adjusted for standard test conditions except the beat frequency oscillator shall be on. The receiver shall be successively tuned to 4.5, 9.0, and 18.0 megacycles in band 1, 2, and 3 respectively. Each test frequency shall be approached from the clockwise and counter-

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clockwise direction, and the difference in dial setting noted (see 3.15.9).

4.8.10 *B.F.O. tuning range.* The receiver shall be adjusted for standard test conditions except the beat frequency oscillator shall be on. The B.F.O. frequency shall be measured at the zero position with a frequency counter. The control shall then be tuned to its maximum positive and maximum negative positions and the frequency measured at both points (see 3.15.10).

4.8.11 *Frequency stability with warm-up.* The receiver shall be adjusted for standard test conditions except the B.F.O. shall be on. After 15 minutes warm-up a frequency counter shall measure the frequency of a signal generator tuned to the receiver frequency. The frequency shall be measured again after three hours. This test shall be performed at 6.0, 12.0, and 24.0 megacycles in band 1, 2, and 3 respectively (see 3.15.11).

4.8.12 *Plate voltage range.* The input voltage shall be adjusted to produce a plate voltage of 100 volts. The signal generator output with modulation shall be adjusted to 10 microvolts output at each test frequency to produce a reference output. The plate voltage shall then be adjusted to 110 volts and 95 volts, and the change in output noted. The test shall be performed at 4.5, 9.0, and 18.0 megacycles (see 3.15.12).

4.8.13 *Overall audio frequency response.* The receiver shall be tuned to 3.0 megacycles under standard test conditions except the gain control shall be adjusted to one half maximum. The signal generator with 30 percent modulation of 1000 cps shall be set to produce a reference output of 2.5 milliwatts. The modulation frequency shall then be varied to 300, 500, 800, 1300, 1600, 2000, 2400, and 3000 cps and the change in output recorded (see 3.15.13).

4.8.14 *Audio power output.* The receiver shall be tuned to 3.0 megacycles under standard test conditions except the gain control shall be set at maximum. The signal generator output with 30 percent modulation of 1000 cps shall be set to 500 microvolts and the power output measured (see 3.15.14).

4.8.15 *Crystal Operation.* The receiver shall be adjusted for standard test conditions except the gain control shall be set at two thirds maximum. Using a 4.8 megacycle crystal, the sensitivity shall be measured at 4.345 and 9.145 megacycles with a signal generator output of 1.5 microvolts modulated. The 4.8 megacycle crystal shall be replaced with a 6.2 megacycle crystal, and the sensitivity shall be measured at 18.145 megacycles with a signal generator output of 3 microvolts modulated (see 3.15.15).

4.9 Transmitter electrical tests.

4.9.1 *Frequency range.* Operation shall be checked at the high and low frequency limits of each band by insertion of the proper crystals (see 3.16.1).

4.9.2 *Power output.* The plate voltage shall be set at 450 volts by adjustment of the input voltage. The power output shall be measured with a 73 ohm load and a 1000 ohm load at 3.0 and 5.5 megacycles in band 1, at 5.5 and 9.0 megacycles in band 2, at 9.0 and 16.0 megacycles in band 3, and at 16.0 and 22.0 megacycles in band 4 (see 3.16.2).

4.9.3 *Keying characteristics.* The keyed output signal shall be observed to determine compliance with 3.16.3.

4.9.4 *Keying speed.* The transmitter shall be keyed at 300 words per minute using the AN/GRA–71( ) Coder-Burst Transmission Group and the received signal recorded. The original message shall be readily intelligible when the recording is played back at $\frac{1}{10}$ to $\frac{1}{20}$ of the recording speed (see 3.16.4).

4.10 Power Supply PP–2684( )/GRC–109 electrical tests.

4.10.1 *Power sources.* Operation from the required power sources shall be tested in accordance with 4.10.2 (see 3.17.1).

4.10.2 *Power outputs.* Equivalent load resistance shall be connected to the power supply. The output voltages shall be measured with the required input voltages to determine compliance with 3.17.2.

4.10.3 *Ripple voltage.* The output ripple voltage using a.c. power inputs shall be meas-

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ured with an electronic voltmeter (see 3.17.3).

4.10.4 *Noise output.* The noise output from vibrator hash with 6 volt storage battery input shall comply with 3.17.4.

4.10.5 *Battery charging output.* The battery charging output using 110 volts a.c. power input shall comply with 3.17.5.

4.11 Power Supply PP-2685( )/GRC-109 electrical tests.

4.11.1 *Power outputs.* Equivalent load resistances shall be connected to the power supply. The output voltages shall be measured with the required input voltages to determine compliance with 3.18.1.

4.11.2 *Ripple voltage.* The output ripple voltage shall be measured with an electronic voltmeter (see 3.18.2).

4.12 Service conditions tests.

4.12.1 *Temperature test.* All units shall be subjected to the temperature cycle shown in Standard MIL-STD-169. For the receiver and the transmitter, the following measurements shall be made alternately with Power Supply PP-2684( )/GRC-109 and Power Supply PP-2685( )/GRC-109 at Steps 1, 3, 8, and 10:

Measurement	*Paragraph*
Sensitivity	4.8.2
Crystal operation	4.8.15
Power output	4.9.2

4.12.2 *Moisture-resistance test.* All units shall be tested as follows:

4.12.2.1 *Test conditions.*

(a) Do not move units from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform the entire test with the covers removed from all units.

4.12.2.2 *Test methods.*

(a) Dry at 130° ±5° F. for 24 hours.

(b) Condition at 77° ±5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.12.2.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.12.2.3 shall be made during the periods shown on the standard.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, all units shall meet full specified performance for those measurements specified in 4.12.2.3.

4.12.2.3 *Performance.* There shall be no mechanical deterioration as evidenced by corrosion of metal parts or binding of rotating parts. The following measurements shall be made on the receiver and the transmitter alternately using Power Supply PP-2685-( )/GRC-109 and Power Supply PP-2685-( )/GRC-109:

Measurement	*Paragraph*
Sensitivity	4.8.2
Crystal operation	4.8.15
Power output	4.9.2

4.12.2.4 *Failure.* If any unit fails to meet the performance specified in 4.12.2.3 during cycling, it shall be realined or readjusted once. If the unit then fails to meet 4.12.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the unit fails to meet full specification requirements after conditioning and adjustment, it does not pass the test.

4.12.3 *Elevation test.* All units, placed in their normal operating position in an altitude chamber, shall be tested as follows (To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment):

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(a) With all units placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Measurement	*Paragraph*
Sensitivity	4.8.2
Power output	4.9.2

(b) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.12.3(a). The units shall meet specified performance for those measurements.

(c) Remove the power from the units and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.12.3(a). The equipment shall meet specified performance for these measurements.

4.12.4 *Immersion test.* All units, with covers removed, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of all units shall be 40° F. or more, above the temperature of the water. The tank in which the units are immersed shall be of sufficient capacity to maintain the water within ±2° F. of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the units shall be removed from the water and wiped dry on exterior surfaces. When the units are opened, there shall be no evidence of leakage.

4.12.5 *Orientation test.* All units shall be inclined a minimum of 5 minutes in each plane (forward, backward, left, and right to an angle of 20 ±3 degrees. During inclination in each plane, the units shall meet the operational test of 4.14.

4.12.6 *Rain test.* All units shall be tested as follows:

4.12.6.1 Dry at 150°±5° F. for 48 hours.

4.12.6.2 Condition for 4 hours at 77°± 5° F. and 40 to 60 percent relative humidity.

4.12.6.3 Take the following measurements to insure that the units meet specified performance.

Measurement	*Paragraph*
Sensitivity	4.8.2
Power output	4.9.2

4.12.6.4 Place the units, in their normal operating position, in a test chamber maintained at 77° ±5° F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5° F. and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.12.6.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blow horizontally at 20 ±5 miles per hour against one side of the units. The water shall fall on exposed surfaces and be blown by the wind against one side of the units. Maintain this condition for 1 hour.

4.12.6.6 Rotate the units 90 degrees horizontally, to expose a second side of the components to the wind and rain. Maintain this condition for 1 hour.

4.12.6.7 Continue as in 4.12.6.6, until all four sides of the units have been exposed to the wind and rain, each side for 1 hour.

4.12.6.8 Immediately after completion of the foregoing, repeat the measurements specified in 4.12.6.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The units shall meet specified performance.

4.12.6.9 Open the units. There shall be no visible evidence of leakage.

4.12.7 *Sand and dust test.* All units shall be tested as follows:

(a) Place the units in their normal operating position, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test

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space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U.S. standard sieve series	Percent
100-mesh	100
140-mesh	98±2
200-mesh	90±2
325-mesh	75±2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition losses	0 to 2

(d) Maintain internal temperature of test chamber at 77° ±4° F. for a period of 6 hours with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After 6 hours at above conditions, raise the temperature to 160°±4°F. and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove units from test chamber and allow them to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the units. However, dust shall not be removed by either blast or vacuum cleaning. The units shall meet specified performance for the following measurements:

Measurement	Paragraph
Sensitivity	4.8.2
Power output	4.9.2

(g) Open the units. There shall be no evidence of deterioration of either external or internal parts.

4.13 Air-seal test. All units shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the units shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the units) applied to the interior of the units by removing the desiccator cartridge. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.14 Operational inspection. Each unit shall be tested for proper operation of all operating controls, switches, meters, and indicator lights. The test frequencies shall be selected by the Government. A continuous wave (cw) signal shall be transmitted and received, and the quality of the received signal observed to determine proper operation of the transmitter, the receiver, and the power supplies.

4.15 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.23. When a listed dimension is not within specific or design limits, it shall be considered a major effect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.16 500 hour operational test (see 3.22). Radio Set AN/GRC-109( ) shall be operated under standard test conditions for 500 hours with the transmiter power output 10 watts at 9.0 megacycles. At the conclusion of the 500 hours, the following tests shall be conducted to determine compliance with 3.22:

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(a) Sensitivity (see 4.8.2).
(b) IF Selectivity (see 4.8.3).
(c) Overall audio frequency response (see 4.8.13).
(d) Audio power output (see 4.8.14).
(e) Power output (see 4.9.2).
(f) Power outputs (see 4.10.2).
(g) Ripple voltage (see 4.10.3).
(h) Power outputs (see 4.11.1).
(i) Ripple voltage (see 4.11.2).

4.17 **Visual and mechanical inspection.** The units shall be examined for the defects in Standard MIL-STD-252.

4.18 **Quality conformance inspection of preparation for delivery.** Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5 below.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.**

5.1.1 *Level A.* Radio Set AN/GRC-109( ) shall be preserved, packaged and tested in accordance withe the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration and damage during worldwide shipment, handling and open storage.

5.1.2 *Level C.* Radio Set AN/GRC-109( ) shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice.

5.2 **Packing.** Radio Set AN/GRC-109( ), packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown unless an individually packed item exceeds that amount. Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, ½-inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping.

5.2.1 *Level A.*

Box specification		*Maximum gross weight*
Fiberboard (Type 1, Class 2)	PPP-B-636	Table VI
Nailed Wood (Style 4, Class 2)	PPP-B-621	200 lbs.
Wirebound (Style 2 or 3, Class 2)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Overseas)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Overseas)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.
Crate, Wood, Sheathed	MIL-C-104	Specification limit
Crate, Wood, Open	MIL-C-3774	"
Crate, Wood, Open	MIL-C-132	"

5.2.2 *Level B.*

Box specification		*Maximum gross weight*
Fiberboard (Type 1, Class 1)	PPP-B-636	Table 1
Nailed Wood (Style 4, Class 1)	PPP-B-621	200 lbs.
Wirebound (Style 3, Class 1)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Domestic)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Domestic)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.
Crate, Wood, Open	MIL-C-3774	Specification limit
Crate, Wood, Open	MIL-C-132	Specification limit

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5.2.3 *Level C.* Radio Set AN/GRC-109( ) shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.2.4 *Pilot pack.* When Level "A" packaging and packing is specified above, one acceptable model of Radio Set AN/GRC-109 ( ) will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Radio Set AN/GRC-109( ) is used to provide medium range continuous wave (cw) communication using a hand key or a high speed, automatic keyer.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction pack(s) as follows:
Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

6.3 Color. The color chip furnished by the procuring agency (3.8) will be color chip No. X-24087 and may be obtained upon request to Commanding General, U. S. Army Material Agency, ATTN: SELMA-J4b, 225 South Eighteenth Street, Philadelphia, Pa. 19103.

6.4 Definitions of inspection terms. The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference:

6.4.1 *Inspection.* Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.1.1 *Examination.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.1.2 *Testing.* Testing consists of determinations of compliance, using technical means.

6.5 Definition of level B. Level B. packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.2).

6.7 Location of operational inspection and air-seal test. It is desirable that the operational inspection (4.14) and the air-seal test (4.13) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment

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which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspection sample units) be sampled and inspected immediately prior to packaging.

6.8 **Nomenclature.** The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-109W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.9 **Government furnished equipment.** A model of Coder-Burst Transmission Group AN/GRA-71( ) should be furnished by the Government for testing purposes.

6.10 **Verification inspection.** Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Project No. 5820-A-209

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding General
U.S. Army Electronics Command
Attn: AMSEL-PP-E-SED
Fort Monmouth, N.J. 07703

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